

8. (a) State the meaning of the term phenotype. [1]

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(b) The ability to roll the tongue into a U-shape is controlled by a dominant allele (**T**).
The recessive allele is represented by **t**.



A man heterozygous for tongue rolling has a child with a woman who is recessive for the characteristic.

(i) Use the letters given above to give the genotype of: [1]

the man

the woman

(ii) Complete the Punnett square below to show the possible genotypes of their children. [2]

(iii) Complete the following to show the **ratio** of the genotypes of the children. [1]

Homozygous dominant : Heterozygous : Homozygous recessive

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- (c) In 1952 Professor Philip Matlock investigated tongue rolling in 33 pairs of genetically **identical** twins. Professor Matlock published the data shown in **Table 8.1**.

Table 8.1

Feature	Number of pairs of identical twins
both can roll their tongues	18
neither can roll their tongues	8
one twin can roll their tongue but the other cannot.	7
Total	33

[Source: Philip Matlock, Ohio State University]

State the evidence from **Table 8.1** which suggests that tongue rolling is not controlled entirely by a gene. [1]

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